



**S. B. JAIN INSTITUTE OF
TECHNOLOGY, MANAGEMENT &
RESEARCH, NAGPUR.**

Practical No. 8

Aim: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

Name of Student:

Roll No.:

Semester/Year:VI/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Understand semantic networks as a knowledge representation technique
- Implement ISA and HAS-A relationships
- Demonstrate inheritance in semantic networks
- Query knowledge programmatically using Python

THEORY:

Semantic Network

A **semantic network** is a graphical knowledge representation technique consisting of:

- **Nodes** → Concepts / Objects
- **Edges** → Relationships

Common relations:

- ISA → Inheritance
- HAS-A → Property
- CAN → Ability

It is closely related to **predicate logic representation**, where:

isa(dog, animal).
has(dog, tail).
can(bird, fly).

Relationship to Predicate Logic

A semantic network can be represented using first-order predicates:

Semantic Relation Predicate Form

Dog is an Animal isa(dog, animal)

Bird canFly can(bird, fly)

Dog has Tail has(dog, tail)

Inheritance rule:

If

$\text{isa}(X, Y)$ and $\text{has}(Y, Z)$

Then

$\text{has}(X, Z)$

Semantic Network Example

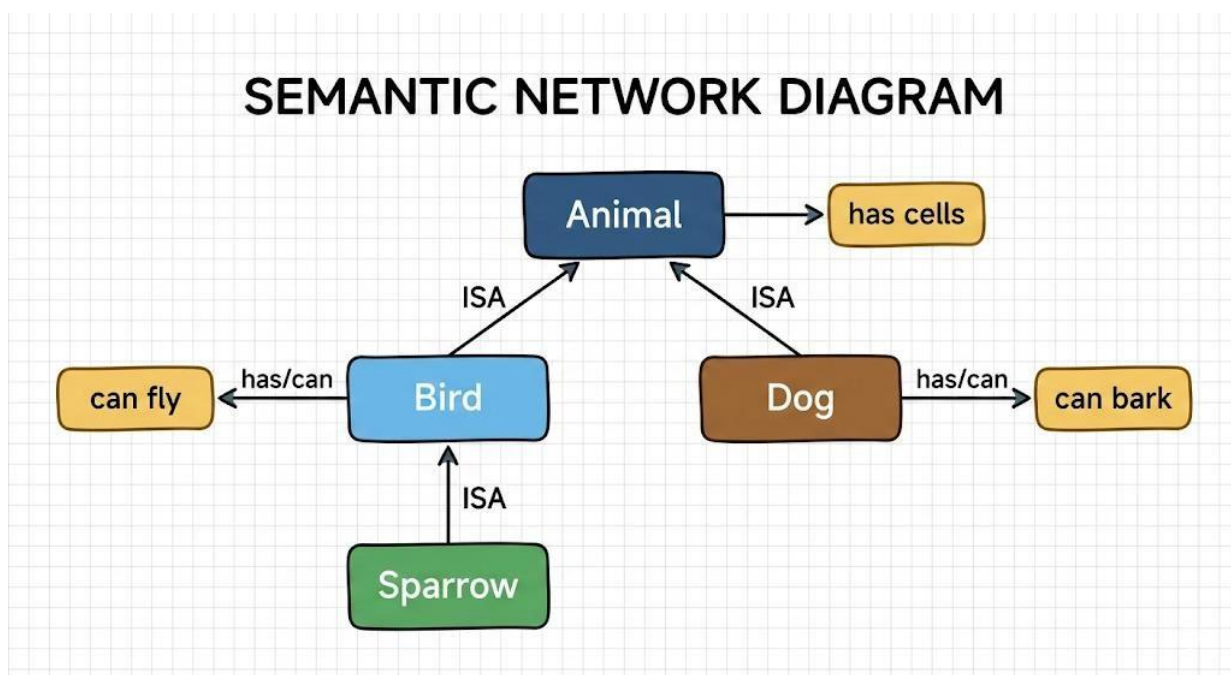
Nodes:

- Animal
- Bird
- Dog
- Sparrow

Relationships:

- Bird ISA Animal
- Dog ISA Animal
- Sparrow ISA Bird
- Animal has cells
- Bird can fly
- Dog can bark

SEMENTIC NETWORK DIAGRAM:



PROGRAM CODE:

```
isa = {
    "bird":"animal",
    "dog":"animal",
    "sparrow":"bird",
}

has_a = {
    "animal" : ["cells"],
    "bird" : ["wings"],
    "dog" : ["tail"],
}

can_do = {
    "bird" : ["fly"],
    "dog": ["bark"],
}

def get_superclass(concept):
    return isa.get(concept, None)

def inherits_property(concept, property_name):
    if concept in has_a and property_name in has_a[concept]:
        return True

    parent = get_superclass(concept)
    if parent:
        return inherits_property(parent, property_name)

    return False

def inherits_ability(concept, ability_name):
    if concept in can_do and ability_name in can_do[concept]:
        return True

    parent = get_superclass(concept)
    if parent:
        return inherits_ability(parent, ability_name)

    return False

print("sparrow has wings:", inherits_property("sparrow","wings"))
print("dog has wings:", inherits_property("dog","wings"))
print("dog has tail:", inherits_property("dog","tail"))

print("sparrow can fly:", inherits_ability("sparrow","fly"))
print("dog can fly:", inherits_ability("dog","fly"))
print("dog can bark:", inherits_ability("dog","bark"))
```

INPUT & OUTPUT:

The screenshot shows the Programiz Python Online Compiler interface. The left pane contains a Python script named `main.py` with the following code:

```

1
2- isa = {
3     "bird": "animal",
4     "dog": "animal",
5     "sparrow": "bird",
6 }
7
8- has_a = {
9     "animal" : ["cells"],
10    "bird" : ["wings"],
11    "dog" : ["tail"],
12 }
13
14- can_do = {
15    "bird" : ["fly"],
16    "dog": ["bark"],
17 }
18
19- def get_superclass(concept):
20     return isa.get(concept, None)
21
22- def inherits property(concept, property name):

```

The right pane, titled "Output", shows the results of the code execution:

```

sparrow has wings: True
dog has wings: False
dog has tail: True
sparrow can fly: True
dog can fly: False
dog can bark: True

```

Below the output, it states: `=== Code Execution Successful ===`

The browser's taskbar at the bottom shows the Windows Start button, task view, and several open applications: Desktop, Firefox, Online Python Co..., and AI Lab Practical no ... The system clock indicates 13:27 on 3/10/2026.

CONCLUSION:**DISCUSSION QUESTIONS:**

1. What is a semantic network?

2. What is ISA relationship?

3. What is inheritance in knowledge representation?

4. How is predicate logic used here?

5. What are advantages of semantic networks?

REFERENCES:

- <https://www.geeksforgeeks.org/artificial-intelligence/semantic-networks-in-artificial-intelligence/>
- <https://thecloudflare.com/knowledge-representation-using-the-semantic-network-in-ai/>
- <https://www.siliconvalleycloudit.com/knowledge-representation-in-ai-semantic-networks/>